

ZISHUO HUANG

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EDUCATION

Department of Electrical Engineering, Tsinghua University

Beijing, China

Ph.D. in Electrical Engineering

2022 - 2027 (*expected*)

- Advisor: Prof. Wenchuan Wu

- Research area: optimization and control of renewable energy sources, especially wind farms

School of Electrical Engineering, Xi'an Jiaotong University

Xi'an, China

B.S. in Electrical Engineering (*Qian Xuesen's Class*)

2018 - 2022

Class of the Gifted Young, Xi'an Jiaotong University

Xi'an, China

2016 - 2018

PUBLICATIONS

1. **Zishuo Huang**, Chenhui Lin, Qi Wang, Wenchuan Wu. Frequency-Routed Coordinated Pitch-Torque Control for Fatigue-Relevant Response Mitigation in Floating Offshore Wind Turbines. *IEEE Transactions on Sustainable Energy*, Early Access, 2026. (IF: 10.3)
2. **Zishuo Huang**, Chenhui Lin, Qi Wang, Wenchuan Wu. A Bilevel Optimization Method for Wind Farm Layout Considering Yaw Control. *IEEE Transactions on Industrial Informatics*, 2025. (IF: 9.8)
3. **Zishuo Huang**, Wenchuan Wu. Dual-Stage MPC-Based AGC for Wind Farm Considering Aerodynamic Interactions. *IEEE Transactions on Sustainable Energy*, 2025. (IF: 10.3)
4. **Zishuo Huang**, Wenchuan Wu, Chenhui Lin, Zizhen Guo. Cooperative Strategies for Frequency Control of Wind Turbines to Mitigate Secondary Frequency Dip: Coefficient Allocation and Exit Techniques. *IEEE Transactions on Sustainable Energy*, 2025. (IF: 10.3)
5. **Zishuo Huang**, Wenchuan Wu. An Efficient Solution for Large Offshore Wind Farm Power Optimization with the Porté-Agel Wake Model: Optimality and Efficiency. *Energy*, 2024. (IF: 10.1)
6. **Zishuo Huang**, Wenchuan Wu, Zizhen Guo. Data-driven Stochastic Model Predictive Control for Regulating Wind Farm Power Generation with Controlled Battery Storage. *IET Renewable Power Generation*, 2023. (IF: 3.2)
7. **Zishuo Huang**, Wenchuan Wu, Hongyan Liu, Jisi Wu, Hao Liu. Real-Time Digital Simulation of DFIG Wind Farm Primary Frequency Control via Droop and Virtual Inertia Control. *6th International Conference on Electrical Technology and Automatic Control (ICETAC)*, 2025.
8. Guojian Li, **Zishuo Huang**, Zhenfeng Wang, Jin Xu, Jihui Dong, Jiang Li. Cooperative Yaw and Induction Control for Wind Farm and Its Effectiveness in Providing Secondary Frequency Regulation. *IEEE 8th Conference on Energy Internet and Energy System Integration (EI2)*, 2024.
9. Guannan Wang, **Zishuo Huang**, Wenchuan Wu, Xiaowei Fan, Jun Zhang, Haitao Liu. Model Predictive Control for Regulating Battery Storage to Smooth the Wind Farm Power Generation. *IEEE International Conference on Power System Technology*, 2023.

PROJECTS	Zhangjiakou 100% Renewable Energy Aggregation and Operational Control: Research & Demonstration <i>Tsinghua University</i> 2023.04 - 2025.07 In this project, I was responsible for modeling and implementing a plant-level active support control scheme. I first proposed a measurement-driven lifted-mapping approach for global linearization, mapping low-dimensional, highly nonlinear processes to a high-dimensional linear state-space model, which produced an interpretable and tractable dynamic-control prototype and a model–data co-driven framework for power tracking and frequency/voltage support; I then extended the prototype to a stochastic MPC with explicit wind-speed forecast-error correction and a deterministic reformulation, unifying in the objective function the penalties on grid-side power fluctuation, schedule-tracking deviation, curtailment, and battery degradation cost, while enforcing engineering constraints on energy storage, pitch angle, and grid voltage to realize wind–storage coordinated optimization. The control performance of the cluster controller was validated via real-time simulation on RTDS.
AWARDS AND HONORS	• National Second Prize (Team Leader), China Undergraduate Mathematical Contest in Modeling (CUMCM) 2019.11 • Provincial First Prize (Shaanxi), China Undergraduate Mathematical Contest in Modeling (CUMCM) 2018.11 • First Prize (Shaanxi Division), 11th National Undergraduate Mathematics Competition 2019.10 • First Prize, Xi'an Jiaotong University Mathematical Modeling Competition 2019.05 • First Prize, Xi'an Jiaotong University Mathematical Modeling Competition 2018.05 • Qiu Changrong Scholarship (Second Class), Academic Year 2019–2020 2020.11 • Qiu Changrong Scholarship (Second Class), Academic Year 2018–2019 2019.11 • Outstanding Student, Qian Xuesen College, Academic Year 2019–2020 2020.10 • Outstanding Student, Qian Xuesen College, Academic Year 2018–2019 2019.10
SKILLS	Languages: Chinese, English. Programming: Python, C++, MATLAB. Simulation softwares: Simulink, RTDS (RSCAD), Powerfactory
ACADEMIC SERVICES	Reviewers for: <i>IEEE Transactions on Sustainable Energy</i>, <i>IEEE Transactions on Industrial Informatics</i>,